# Gov51 Project

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https://github.com/angelacalderonpio/finalproject

I am submitting this one day late (on 5/12) as I received a 24-hour extension from Professor Cunningham due to my computer unfortunately not working.

## Introduction

Education is often treated as one of the clearest ways to expand opportunity (Duncan and Murnane), particularly for children of low-income backgrounds. The mechanism, a straightforward one, dictates that schools that spend more per-pupil will have with more resources, which in turn better supports students. The implication in this mechanism is that stronger schools should help students move beyond the economic position they were born into. This idea is especially important for low-income students, since public education is often perceived as one of the main institutions through which the connection between family background ans adult outcomes is weakened.

But this raises an empirical question: do schools that spend more per pupil *actually* show more economic mobility?

This paper examines how strongly per-pupil education spending is **associated** with intergenerational mobility across U.S. commuting zones.

To answer this question, I combine data from Opportunity Insights on intergenerational mobility with per-pupil spending data from the National Center for Education Statistics. The analysis uses commuting zones as the unit of observation, allowing comparisons across local labor-market areas throughout the United States.

I expect the relationship between spending ans mobility to be complicated. A simple opportunity-based argument would predict that higher-spending places should have greater mobility, because more school resources should help weaken the connection between family background and adult incomes. However, this expectation may not hold if high-spending places are also places where family advantages are strongly reproduced, say, a wealthy area that emphasizes local taxes as a means to fund education, a practice commonplace in Texas, for example. For that reason, this paper treats the relationship as an empirical question as opposed to assuming that higher spending automatically corresponds to more mobility.

I treat this as a descriptive study rather than a causal one: the goal isn't to prove that spending causes mobility, but ask whether higher-spending places tend to show lower income persistence across generations. The objective is to lend evidence in the body of literature surrounding this matter. The main contribution if this paper is therefore measurement and interpretation: it uses descriptive statistics, visualizations, correlation, and. regression to evaluate where spending and mobility move together across commuting zones.

The results suggest that higher per-pupil spending is associated with slightly greater mobility, but the relationship is tenuous. The scatterplot shows a dispersed pattern rather than a clear trend (Figuere 1), the spending-group graph (Figure 2) does not show consistent improvement as spending eises, and the regression coefficient is statistically significant but small. Overall, the evidence suggests that spending may matter, but spending alone does not explain why some places show more mobility than others.

## Data

This project combines two main sources of data. The first is Opportunity Insights data on intergenerational mobility. The second is the National Education for Education Statistics (NCES) data on per-pupil education spending. The first dataset involves a cohort of children born in the early 1980s, and their family's income versus their incomes as adults. The second dataset is per-pupil education spending from the 1989-90 school year, a year that would be relevant to the children of the first dataset. After merging the datasets, the unit of observation is the commuting zone, resulting in a sample of 381 commuting zones.

The outcome variable is the rank-rank slope. This variable measures how strongly a child's adult income rank is associated with their parent's income rank. Higher rank-rank slopes indicate greater income persistence across generations, meaning that parental income is more predictive of where children end up economically as adults. Lower rank-rank slopes indicate greater mobility because children's adult outcomes are less tied to their parents' income rank.

The main independent variable is per-pupil education spending. Because the spending data are measured at the state level, each commuting zone is assigned the spending value of its state. This allows for broad comparison across places, although it introduces an important limitation because spending can vary within states.

Region is also included in visualizations in order to compare patterns across different parts of the country.

## Analysis

The analysis examines the relationship between per-pupil education spending and intergenerational mobility in four ways. First, I use a scatter plot to visualize the overall relationship between spending and the rank-rank slope. Second, I group commuting zones into spending categories and compare their average and median mobility outcomes. Third, I calculate the correlation between spending and mobility. Finally, I estimate a simple OLS regression of rank-rank slope on spending.

Because this is a descriptive project, the regression results should be interpreted as **associations rather than causal effects**.

### Set-up

### Summary Statistics

Table 1: Summary Statistics (Table 1)

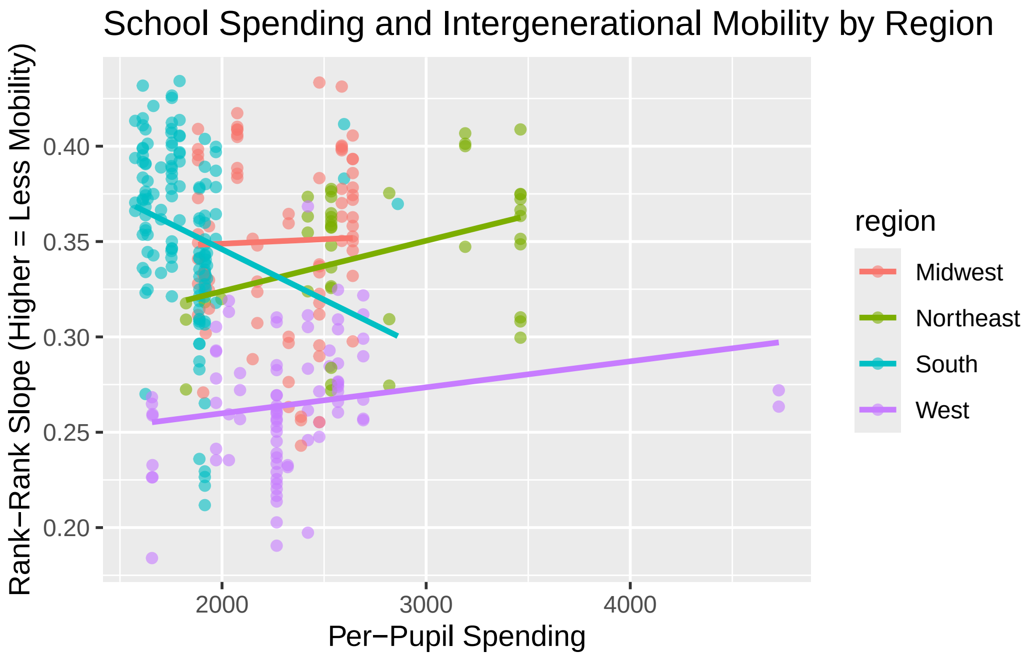
| Mean Spending | SD Spending | Min Spending | Max Spending | Mean Mobility | SD Mobility | Min Mobility | Max Mobility | Observations |
|---:|---:|---:|---:|---:|---:|---:|---:|---:|
| 2170.93 | 478.59 | 1574.44 | 4727.59 | 0.33 | 0.06 | 0.17 | 0.43 | 381 |

Table 1 reports summary statistics for per-pupil spending and intergenerational mobility across commuting zones, the two main variables. On average, per-pupil spending is approximately $2,170.93, with variation (SD is about $478.59), ranging from $1,574.44 to $4,727.59.

The mean rank-rank slope is 0.33, suggesting a moderate degree of persistence in income across generations. Since higher values indicate less mobility, this means that parent income remains meaningfully associated with child income outcomes in the average commuting zone.

The table also demonstrates substantial variation across commuting zones, with values ranging from 0.18 to 0.43 in rank-rank slope. This variation is important because the project depends on comparing education spending levels and different mobility outcomes

**Figure 1**




```
Call:
lm(formula = rank_rank_slope ~ spending, data = df_merged)

Residuals:
      Min        1Q    Median        3Q       Max 
-0.154840 -0.040887  0.004058  0.045877  0.107510 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept)  3.651e-01  1.419e-02  25.730   <2e-16 ***
spending    -1.582e-05  6.382e-06  -2.478   0.0137 *  
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.05574 on 332 degrees of freedom
Multiple R-squared:  0.01816,	Adjusted R-squared:  0.0152 
F-statistic:  6.14 on 1 and 332 DF,  p-value: 0.01371
```

Figure 1 shows the relationship between per-pupil spending and the rank-rank slope across commuting zones. If higher spending were strongly associated with greater mobility, the points would show a clearer downward trend (because lower rank-rank slopes indicate more mobility). Instead, the overall pattern appears to slope slightly upward, meaning that higher spending is associated with a higher rank-rank slope. Since higher rank-rank slopes indicate less mobility, this goes against the would expectation that higher spending would be associated with greater mobility.

At the same time, the relationship is not especially strong: points are dispersed, and commuting zones with similar spending levels offen have different rank-rank slopes. So, it seems that spending alone does not explain much of the variation in mobility outcomes.

The regional trend lines also suggest that the relationship differs across parts of the country rather than following one uniform national pattern. And while the slopes of the two variables are positive (with the exception of the South, interestingly) they do so at varying strengths.

This figure is important because it complicates the initial intuition behind the project. As aforementioned, a scatterplot confirming my initial impulse (that higher spending is associated with lower rank-rank slopes), we'd be seeing a much different picture. Instead, the graph might suggest that higher-spending places may also be places where income advantages are more persistent across generations. That does not mean spending causes lower mobility, but perhaps suggests that spending may be correlated with other features of wealth that shape mobility.

**Figure 2**

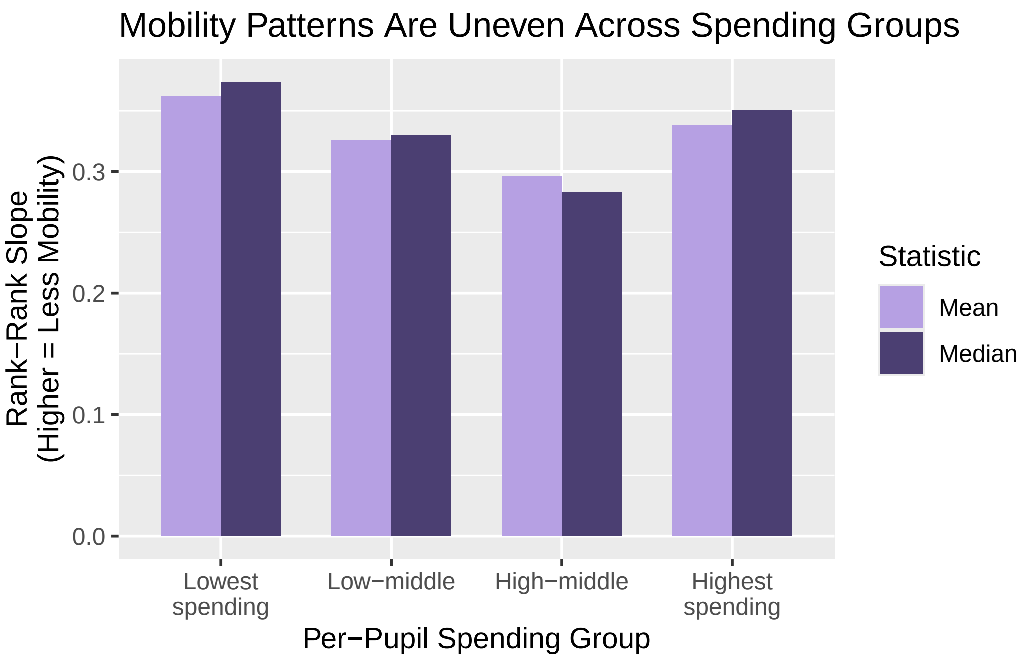



Figure 2 compares the mean and median rank-rank slope across four per-pupil spending groups. I grouped commuting zones into quartiles based on per-pupil education spending. In practice, this means that the "lowest spending" group contains the bottom quart of commuting zones by spending, while the the "highest spending" group contains the top quarter. I include median because averages can be affected by skew or extreme values, so the median provides a better sense of the typical commuting zone within each group.

The graph yields mixed results. While the first three bars seem to match the simple hypothesis that higher spending is strongly associated with lower rank-rank slopes, it is important to note that the highest-spending group has a relatively high rank-rank slope, meaning less mobility. Thus, it suggests that high spending does not necessarily mean that children's adult outcomes are less tied to their parents' income.

One possible interpretation is that some high-spending areas are also places where family advantages matter. That is, wealthier communities may also spend more on education, but they may also have housing markets, school boundaries, social networks, and labor-market advantages that help children from higher-income families remain high-income adults. This emphasizes the role of social capital that the literature surrounding this broader topic hopes to expand towards other income groups in order to bolster adult incomes (Chetty et. al). So, high spending can coexist with high income persistence. Again, this graph supports a broader point that spending alone isn't enough to explain mobility outcomes.

## Correlation

```
[1] -0.1347513
```

The correlation between per-pupil spending and the rank-rank slope is -0.135. The negative sign points in the expected direction. Since lower rank-rank indicates greater mobility, the result suggests that higher spending per-pupil is associated with slightly greater mobility.

However, the magnitude of the relationship is small. A correlation of -0.135 indicates only a weak association between spending and mobility. This reiterates the earlier visual evidence from the scatterplot and spending-group graph, both of which showed a dispersed relationship.

## Linear Regression

Table 2: Regression of Rank-Rank Slope on Per-Pupil Spending with Region Controls

| Variable | Estimate | Std. Error | t value | p value |
|---|---|---|---|---|
| Constant | 0.3307 | 0.0158 | 20.9683 | 0.0000 |
| Per-Pupil Spending ($1,000s) | 0.0085 | 0.0065 | 1.3045 | 0.1930 |
| Northeast | -0.0100 | 0.0085 | -1.1814 | 0.2383 |
| South | 0.0099 | 0.0068 | 1.4575 | 0.1459 |
| West | -0.0861 | 0.0066 | -13.0003 | 0.0000 |

Table 2 reports a simple linear regression of rank-rank slope on per-pupil spending, measured in thousands of dollars. Measuring spending in thousands makes the coefficient easier to interpret substantively.

The coefficient on spending is negative, meaning that higher per-pupil spending is associated with lower rank-rank slopes. Specifically, each additional $1,000 in per-pupil spending is associated with about a 0.0158 decrease in the rank-rank slope.

Although the coefficient is statistically significant, the size of the relationship remains modest. The regression therefore supports the broader conclusion of the paper: spending is associated with mobility in the expected direction, but the relationship is relatively weak and does not explain most of the variation in mobility across commuting zones.

## Conclusion and Limitations

This analysis has several important limitations. First, the project is descriptive and cannot establish causality. The results show that spending and mobility are associated, but they do not demonstrate that higher spending causes mobility to increase. Higher spending places may differ from lower-spending places in many other ways, including housing conditions, local labor markets, and family income levels, to name a few.

Second, the spending variable is measured at the state level and assigned to commuting zones. Because commuting zones and state education systems do not perfectly overalp, the spending measure may not fully capture the actual educational resources available in each commuting zone.

Third, per-pupil spending measures the amount of money spent, but not how those funds are used. In other words, two places may spend similar amounts per student while providing very different educational experiences.

## Conclusion

This paper examined whether higher per-pupil education spending is strongly associated with intergenerational mobility across U.S. commuting zones. The findings suggest that the relationship exists, but it is weak.

Higher spending is associated with slightly lower rank-rank slopes, meaning slightly greater mobility. However, the scatterplot remains widely dispersed, the spending-group analysis does not show a steady trend, and the regression coefficient is relatively small. These results suggest that spending alone does not explain why some places present greater economic mobility than others.

This does not mean education spending is unimportant. Rather, the findings suggest that spending is likely only one part of a larger opportunity structure that includes schools, neighborhoods, labor markets, and local policy environments.

A future version of this project could use district-level spending rather than state-level measures and include additional controls for poverty, segregation, housing costs, and labor-market conditions. That would help make it easier to determine whether the weak relationship found here reflects the limits of spending itself or the influence of wider local conditions.

In conclusion, the most important takeaway is that the relationship runs against the simplest version of the spending argument. The results suggest that the question is not simply whether places spend more, but what *kinds* of places spend more and whether those resources reach students in ways that weaken inherited advantage.

## References

Murnane, Richard J. and Greg J. Duncan. Whither Opportunity? Rising Inequality, Schools, and Children's Life Chances. 1 ed. Russell Sage Foundation, 2011. Project MUSE, https://muse.jhu.edu/book/26993.

Raj Chetty et al., Opportunity Insights, "Changing Opportunity: How Changes in Children's Social Environments Have Increased Class Gaps and Reduced Racial Gaps in Economic Mobility," (July 2024).

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